

A FAIRNESS QUESTION FOR TWO FAMILIES OF INVOLUTIONS ON LATIN SQUARES

SETTING

Let L be a Latin square of order n on the symbol set $[n]$, with rows and columns indexed by $[n]$. Two rows, 1 and 2, and two columns, $j \neq j'$, are distinguished throughout. The frame. Let $\pi = \pi_L$ be the permutation of the rows defined by

$$L(\pi(r), j') = L(r, j) \quad (r \in [n]),$$

i.e. $\pi(r)$ is the row in which column j' holds the symbol that column j holds in row r ; we call π the *frame* of L . It is a derangement, since $\pi(r) = r$ would force $L(r, j) = L(r, j')$. The *bit* of L is

$$\beta(L) = \mathbf{1}\{\text{rows 1 and 2 lie in a common cycle of } \pi_L\},$$

and we write C_1, C_2 for the cycles of π_L through rows 1 and 2 (so $\beta = 1$ iff $C_1 = C_2$).

Row-pair permutations. For rows $x \neq y$ let $\rho_{x,y}$ be the permutation of the columns defined by $L(y, \rho_{x,y}(q)) = L(x, q)$. On the columns of any cycle Q of $\rho_{x,y}$ the two rows carry the same set of symbols.

Two families of moves.

- (TRADE) For rows $x \neq y$ with $x, y \notin \{1, 2\}$, let Q be the cycle of $\rho_{x,y}$ containing j' , and exchange the entries of rows x and y in the columns of Q .
- (TURN) For a cycle c of π_L containing neither row 1 nor row 2, exchange the entries of columns j and j' in the rows of c .

Lemma 1. *Both moves produce Latin squares, are involutions, and leave rows 1 and 2 unchanged. A turn replaces π by π^{-1} on the rows of c and leaves it unchanged elsewhere; the cycle partition is unchanged. A trade at (x, y) replaces π by $(xy) \circ \pi$ if $j \notin Q$ (so the cycles of x and y are merged if distinct and split if equal), and by $(xy) \circ \pi \circ (xy)$ if $j \in Q$ (the cycle partition is unchanged up to exchanging the labels x, y). In particular a trade can change β only when $j \notin Q$; call such a pair (x, y) flippable. If $\pi(x) = y$ or $\pi(y) = x$ then (x, y) is not flippable.*

The proofs are direct verifications. Both rows carry the same symbols on Q , and both columns carry the same symbols on c , which gives the Latin property; the traded cycle is again the $\rho_{x,y}$ -cycle of j' in the new square, and the turned cycle is again a cycle of the new frame, which gives the involution property. The frame formulas follow from the definition of π , since only the entries of rows x, y in columns j, j' matter. For the last claim: with $b' = L(y, j')$ and q^* the column in which row x holds b' , one has $\rho_{x,y}(q^*) = j'$, so Q is the $\rho_{x,y}$ -cycle of q^* . If $\pi(x) = y$ then $b' = L(x, j)$, so $q^* = j \in Q$; if $\pi(y) = x$ then $L(x, j') = L(y, j)$, so $\rho_{x,y}(j') = j \in Q$.

COMPONENTS AND THE QUESTION

Fix a $2 \times n$ Latin rectangle R (the first two rows) and columns $j \neq j'$ such that

$$(1) \quad R(1, j) \neq R(2, j') \quad \text{and} \quad R(1, j') \neq R(2, j),$$

i.e. $\pi(1) \neq 2$ and $\pi(2) \neq 1$ for every extension of R . (Without (1) the question below is void: if $R(1, j) = R(2, j')$ then $\pi(1) = 2$, hence $\beta = 1$, for every extension of R .) Let $S(R)$ be the set of Latin squares whose first two rows are R . Each move is an involution of $S(R)$; let $\mathcal{K}(R)$ be the set of orbits (“components”) of the group they generate, and for a component K let

$$p(K) = \frac{|\{L \in K : \beta(L) = 1\}|}{|K|}.$$

We call the bit *fair* on K if $p(K)$ is close to $\frac{1}{2}$.

Question 1 (component fairness). *Is it true that, uniformly over R and over column pairs (j, j') satisfying (1),*

$$\frac{1}{|S(R)|} \sum_{K \in \mathcal{K}(R)} |K| (2p(K) - 1)^2 \rightarrow 0 \quad (n \rightarrow \infty)?$$

Equivalently: for L uniform in $S(R)$, the conditional probability that $\beta(L) = 1$ given the component of L is $\frac{1}{2} + o(1)$ in L^2 . A quantitative version asks for the left side to be $o(1/\log^2 n)$. A weaker, averaged version that is still useful is stated in Remark 3(i).

Remark 1 (why the question is natural). Consider the Markov chain on $S(R)$ which, with probability $\frac{1}{2}$, picks a uniform pair $x, y \notin \{1, 2\}$ and trades, and otherwise picks a uniform row $x \notin \{1, 2\}$ and turns its π -cycle if that cycle avoids rows 1, 2 (holding if not). Each move is an involution chosen with state-independent probability, so the kernel is symmetric and the uniform measure on each component is stationary. Every state has a hold: by (1) the cycle C_1 is not $(1\ 2)$, so it contains a row $x \notin \{1, 2\}$ whose turn is rejected. Hence the chain is aperiodic on each component, and for the stationary chain started from the uniform measure on $S(R)$ the left side of the Question equals $\lim_{t \rightarrow \infty} \mathbb{E}[\beta'(L_0)\beta'(L_t)]$ with $\beta' = 2\beta - 1$, the long-time autocorrelation of the bit.

Along the chain the frame π performs the walk of left multiplication by transpositions (xy) , $x, y \notin \{1, 2\}$, *thinned* to the flippable pairs: the turns reverse a cycle and the non-flippable trades conjugate by (xy) , and neither changes β . The unthinned walk preserves $a = \pi^{-1}(1)$ and $b = \pi^{-1}(2)$. Contracting the arcs $a \rightarrow 1 \rightarrow \pi(1)$ and $b \rightarrow 2 \rightarrow \pi(2)$ identifies the fibre $\{\pi : \pi^{-1}(1) = a, \pi^{-1}(2) = b\}$ with the permutations of $[n] \setminus \{1, 2\}$ having no fixed point outside $\{a, b\}$ (a fixed point at a or b is the 2-cycle $(1\ a)$ or $(2\ b)$ of π), and the walk with the walk by uniformly random transpositions of $[n] \setminus \{1, 2\}$ in which the moves that would create a fixed point — exactly the π -adjacent pairs — are rejected. Rows 1, 2 share a cycle of π iff a, b share a cycle of the contracted permutation, which is the case for exactly half of each fibre: left multiplication by (ab) is an involution of the fibre that toggles it. So if every non-adjacent pair were flippable the answer would be yes, with the left side $o(1)$ after $O(n \log n)$ steps if, as one expects by comparison with the unrestricted walk (Diaconis–Shahshahani), the rejections do not slow the walk down. The content of the Question is that the thinning by flippability does not destroy this.

Remark 2 (what is known numerically). The chain of Remark 1, modified to reject non-flippable trades (its components are finer than those above, so the evidence transfers), was run on uniformly random Latin squares of order $n = 30, 50, 100$, with (j, j') at distance

at least 2 in both directions inside a cycle of the column permutation σ of R defined by $R(2, \sigma(q)) = R(1, q)$. A uniformly random pair $x, y \notin \{1, 2\}$ that is not π -adjacent (i.e. $\pi(x) \neq y, \pi(y) \neq x$) is flippable with probability 0.500 ± 0.001 , the same whether x, y lie in one cycle of π or in two; the π -adjacent pairs, which are never flippable, are a fraction $\approx 2/n$ of all pairs. The autocorrelation of the bit, as a function of the number of trades performed, decays at least as fast as that of the unthinned walk and reaches the noise floor (≈ 0.01) within 64 trades at each n . Column pairs with j, j' in different cycles of σ were not tested at these sizes.

The components have been enumerated exactly for $n \leq 7$ (seven instances at $n = 7$, $|S(R)| \approx 6.6 \cdot 10^6$ each, all cycle types of σ). At $n = 7$ there is always one component containing 99.1–99.7% of $S(R)$, on which $p(K)$ agrees with $\mathbb{P}[\beta = 1 \mid R]$ to four decimals (0.493–0.502 for within-cycle marks, exactly $\frac{1}{2}$ for cross-cycle marks); the remaining squares lie in hundreds of components of size 120–960 and are the squares in which all, or all but one or two, row pairs $\rho_{x,y}$ are single 7-cycles (isotopes of $\mathbb{Z}_7$ and other row-Hamiltonian squares), so that every trade swaps two whole rows and nothing is flippable. These contribute 0.003–0.008, essentially the whole left side. ($n = 6$ is degenerate: the moves preserve the row parity of L there, because all free cycles are 2-cycles.) So the component structure is expected to be trivial for large n — one component up to a super-polynomially small set of locked squares — and the Question with R fixed is then the quenched statement $\mathbb{P}[\beta = 1 \mid R] = \frac{1}{2} + o(1)$.

Remark 3 (weaker forms that are still useful). (i) The application in Remark 4 needs only an averaged version: for every cycle type λ of σ , with L uniform among the Latin squares of order n whose column permutation σ has cycle type λ , R its first two rows, and $(j, j') = (j, \sigma^\alpha(j))$ with j in an m -cycle of σ and $\alpha, m - \alpha \geq 2$ prescribed, the expectation of the left side of the Question is $o(1/\log^2 n)$. (ii) A natural route is to show that, for all but an $o(1/\log^2 n)$ -fraction of $L \in S(R)$, a positive proportion of the bit-changing pairs are flippable — when $\beta = 1$ these are the pairs $x, y \in C_1$ such that rows 1 and 2 lie on different arcs of the split of C_1 into $(x, \pi(x), \dots, \pi^{-1}(y))$ and $(y, \pi(y), \dots, \pi^{-1}(x))$; when $\beta = 0$ they are the pairs $x \in C_1, y \in C_2$ — and to combine this with a mixing estimate for the thinned walk.

Remark 4 (provenance). The Question arose in a programme on the Cavenagh–Greenhill–Wanless conjecture on the cycle type of the second row of a random Latin square. Let $\mathcal{C}_n(\nu)$ be the number of Latin squares of order n whose column permutation σ (defined in Remark 2) has cycle type ν , let λ contain a part $m = \alpha + \alpha'$ with $\alpha, \alpha' \geq 2$, and let μ be λ with that part split into α, α' . For L uniform among the squares of type λ and (j, j') a uniform pair $\{j, \sigma^\alpha(j)\}$ with j in an m -cycle of σ , the marked-pair identity

$$2 \frac{\mathcal{C}_n(\lambda)}{\mathcal{C}_n(\mu)} - 1 = \frac{\mathbb{P}[\beta = 1]}{\mathbb{P}[\beta = 0]}$$

reduces the conjecture to $\mathbb{P}[\beta = 1] = \frac{1}{2} + o(1/\log n)$ over such instances, and a positive answer to the quantitative Question, averaged as in Remark 3(i), gives exactly that. The question itself makes no reference to this.

Remark 5 (an exact partial answer). Put $K_0 = \min(|C_1|, |C_2|)$ if $\beta = 0$ and $K_0 = \min(d_{12}, d_{21})$ if $\beta = 1$, where d_{12}, d_{21} are the numbers of π -steps from row 1 to row 2 and back, and let $(x_k, y_k) = (\pi^{-k}(1), \pi^{-k}(2))$, $1 \leq k < K_0$ (the *ladder*: $2(K_0 - 1)$ distinct rows avoiding 1, 2). The trades at the ladder pairs are commuting involutions of $S(R)$; each preserves the ladder and toggles β iff its pair is flippable. On each orbit of the group they generate, β is therefore

exactly fair unless no ladder pair is flippable, and

$$\mathbb{P}[\beta = 1 \mid R] - \mathbb{P}[\beta = 0 \mid R] = \mathbb{P}[\beta = 1, \text{Flip} = \emptyset \mid R] - \mathbb{P}[\beta = 0, \text{Flip} = \emptyset \mid R],$$

where Flip is the set of flippable ladder indices. So $|2\mathbb{P}[\beta = 1 \mid R] - 1| \leq \mathbb{P}[\text{Flip} = \emptyset \mid R]$, and since the components are essentially everything (Remark 2), the Question reduces to showing that the $K_0 - 1$ ladder pairs are not all crossed except with probability $o(1)$ (or $o(1/\log n)$ for the quantitative version). Numerically $\mathbb{P}[\text{Flip} = \emptyset \mid K_0] = 2^{-(K_0-1)}$ to the resolution of 9000 instances at $n = 30$ and 7200 at $n = 50$. The identity was verified exactly by enumeration at $n = 7$.